

CAPM-Based Tangent Portfolio Optimization and Forecasting Analysis

Introduction: Context and Objective

This exercise applies the Capital Asset Pricing Model (CAPM) framework on real financial history over a 5-year time horizon on Microsoft (MSFT), Johnson & Johnson (JNJ), SPDR Gold Shares (GLD), and Vanguard FTSE Developed Markets ETF (VEA). These securities were selected to include asset class variety (equity vs. commodity), geographic reach (U.S. vs. international), as well as sector (tech, pharma, precious metals). The intention was less about demonstrating predictability, as so much as to see how well the CAPM framework does with and without perfect foresight, as well as how well it predicts under shifting market regimes.

It addresses key issues such as construction of tangent portfolio, security market lines, estimation of beta through regression as well as covariance methods, estimation of CAPM returns, and alpha estimation. It paints an idea of how the unsystematic risk is handled by diversification as well as where the model is limited.

Portfolio Construction and the Capital Market Line

The project began with constructing 50 random portfolios from the four selected securities. Weights were randomly assigned, not with the goal of optimization, but to illustrate the shape of the opportunity set and efficient frontier in risk-return space. Each portfolio's annualized return and risk (standard deviation) over the period April 2020 – March 2023 was computed from monthly returns.

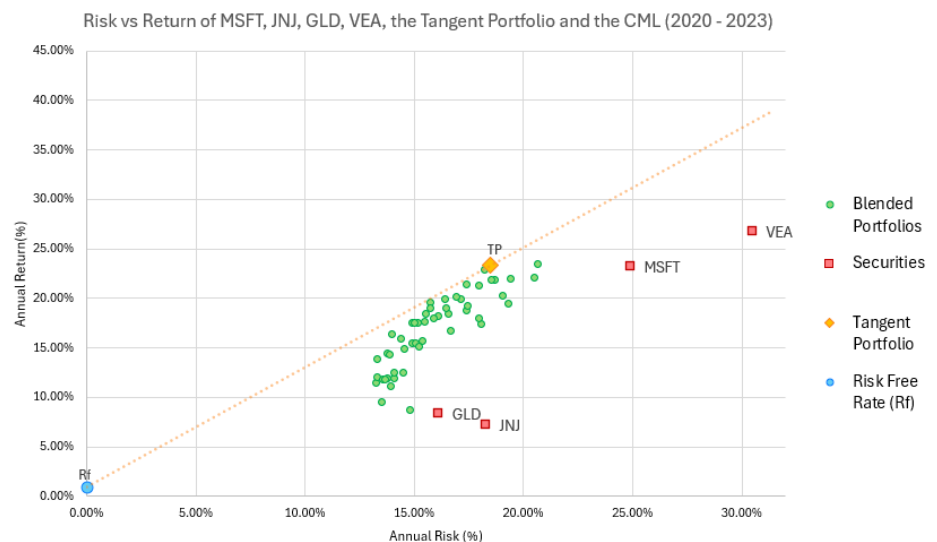
To identify the tangent portfolio (TP), Solver was used to maximize the Sharpe ratio (the slope of the line connecting the risk-free rate of 1% to the TP) under the constraint that the portfolio weights sum to 1.

Interestingly, Solver was able to reach a valid solution without requiring bounds like -30% on short selling, indicating the chosen securities were not extreme outliers in terms of being terrible for returns.

The optimal tangent portfolio weights were:

Security	MSFT	JNJ	GLD	VEA
Weight	50.85%	-0.46%	9.49%	40.13%

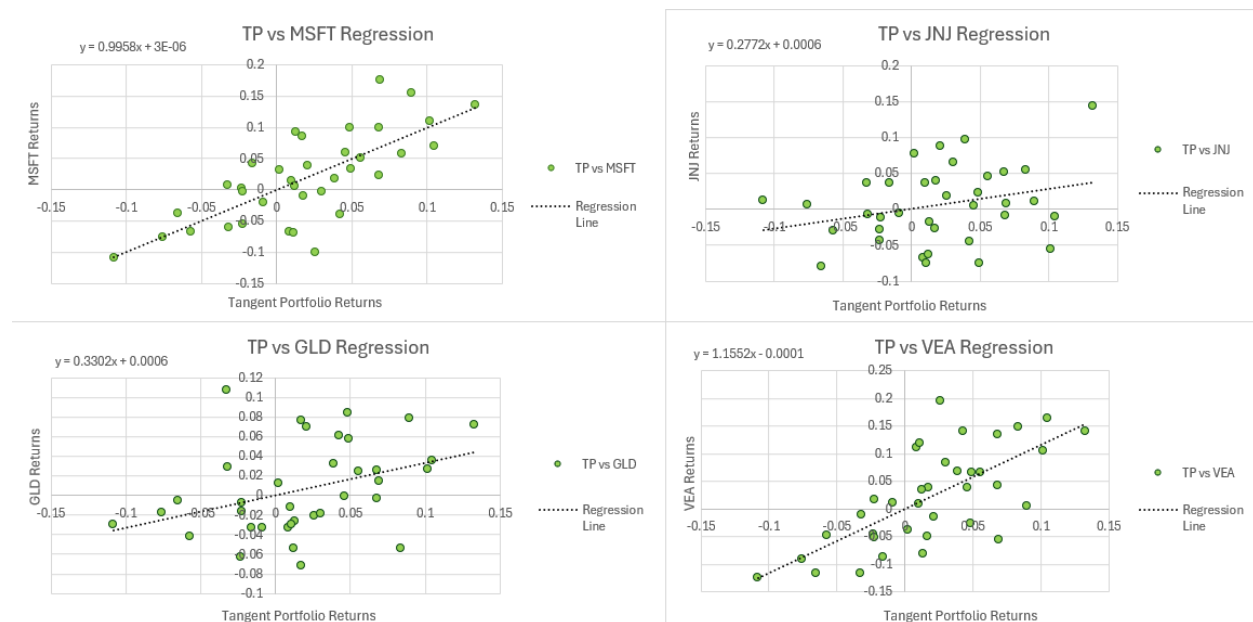
The negative weight for JNJ suggests that, from a theoretical standpoint, the portfolio benefits from shorting JNJ slightly due to its relatively lower historical performance during the post-COVID recovery.



The tangent portfolio yielded an annual return of 23.31% and risk of 18.48%. The Capital Market Line (CML), representing portfolios formed by combining the risk-free asset with TP, was plotted through this point. All 50 portfolios lie below the CML, consistent with theory.

Regression-Based Betas and the Security Market Line

To compute each security's beta relative to TP, monthly returns from April 2020 to March 2023 were regressed against TP returns. Each regression was accompanied by a trendline on a scatterplot with TP on the x-axis and the securities return on the y-axis.



The same betas were confirmed using the covariance formula:

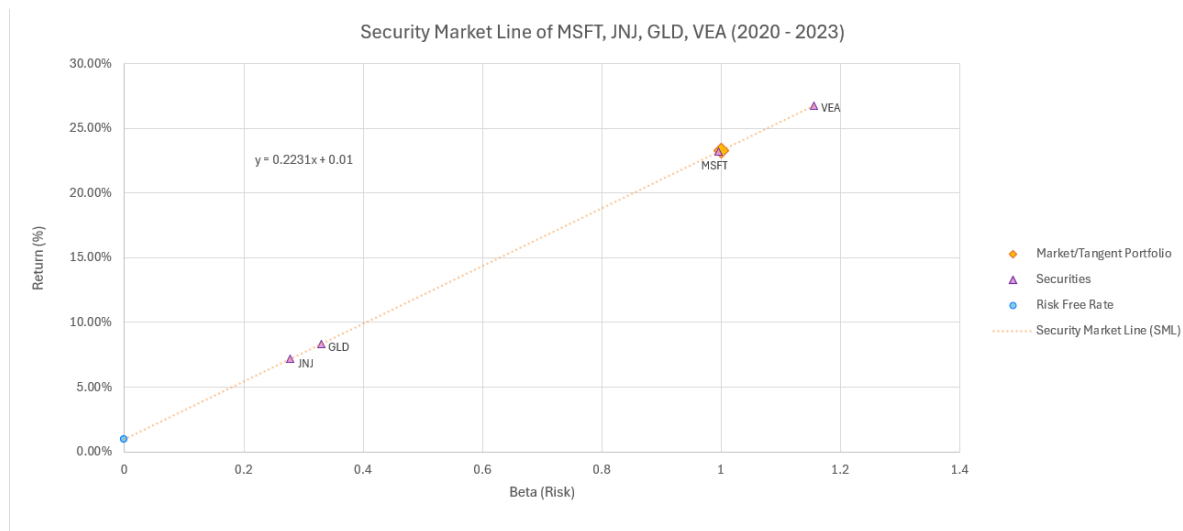
$$\beta_i = \frac{\text{Cov}(R_i, R_{TP})}{\text{Var}(R_{TP})}$$

The consistency of both methods validates their equivalency under the same data inputs.

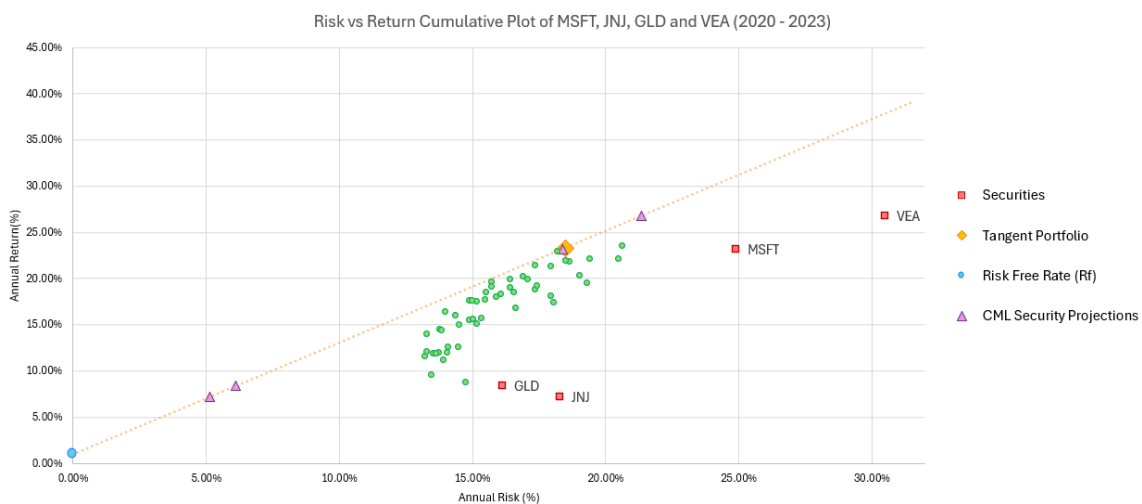
Security	MSFT	JNJ	GLD	VEA
Beta	0.9958	0.2772	0.3302	1.1552

These values reflect the degree of systematic risk (with respect to TP) each asset carries. Regression plots showed moderate spread, likely implying R^2 values between 0.3–0.7. This shows that while TP captures some variation in returns, considerable idiosyncratic noise remains.

The Security Market Line (SML) for 2020–2023 was plotted using these betas, with actual returns placed against their theoretical CAPM expectations. As expected, the points fit nearly perfectly along the SML. See the graph on the next page.



A corresponding CML plot was also made where each security was projected onto the CML through its beta-based mini-portfolio equivalent, and what can be seen clearly is that the mini-portfolios essentially eliminate all the unsystematic risk that each of the original securities have.



Forecasting with CAPM: Post-2023 Market Behaviour

To evaluate the forecasting ability of CAPM, the TP weights obtained from the 2020–2023 period were held constant, and the model was applied to returns from April 2023 to March 2025. Forecasted returns for each security were computed using:

$$E[r_i] = R_f + \beta_i \cdot (E[r_{TP}] - R_f)$$

With a risk-free rate of 1% and TP annual return over the next two years of only 0.52%, the forecasted CAPM returns were remarkably low:

Security	MSFT	JNJ	GLD	VEA
Forecasted Return	0.52%	0.87%	0.84%	0.45%

In contrast, the actual returns were much higher (or lower), revealing large alphas:

Security	MSFT	JNJ	GLD	VEA
Actual Return	14.87%	4.64%	23.64%	-23.07%
Alpha	14.34%	3.78%	22.80%	-23.52%

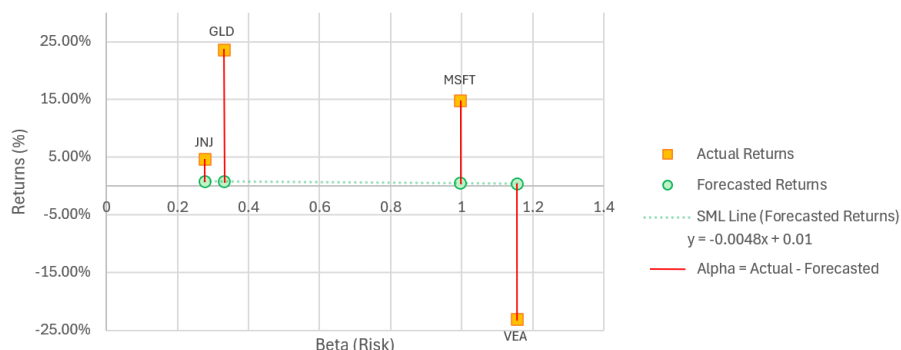
The substantial deviations in returns clearly illustrate that CAPM's predictive performance without reoptimization deteriorates significantly over time. This occurs because the Old TP used to predict this forecast was optimized for how the securities were returning before, and it does not consider the fact that these securities returns changed significantly in 2023-2025. What was specifically interesting was VEA, which experienced a sharp -23%

alpha. Meanwhile, GLD had a strong positive alpha, perhaps reflective of macroeconomic shifts toward “safe haven” assets amid rising fear and uncertainty, especially when considering current events.

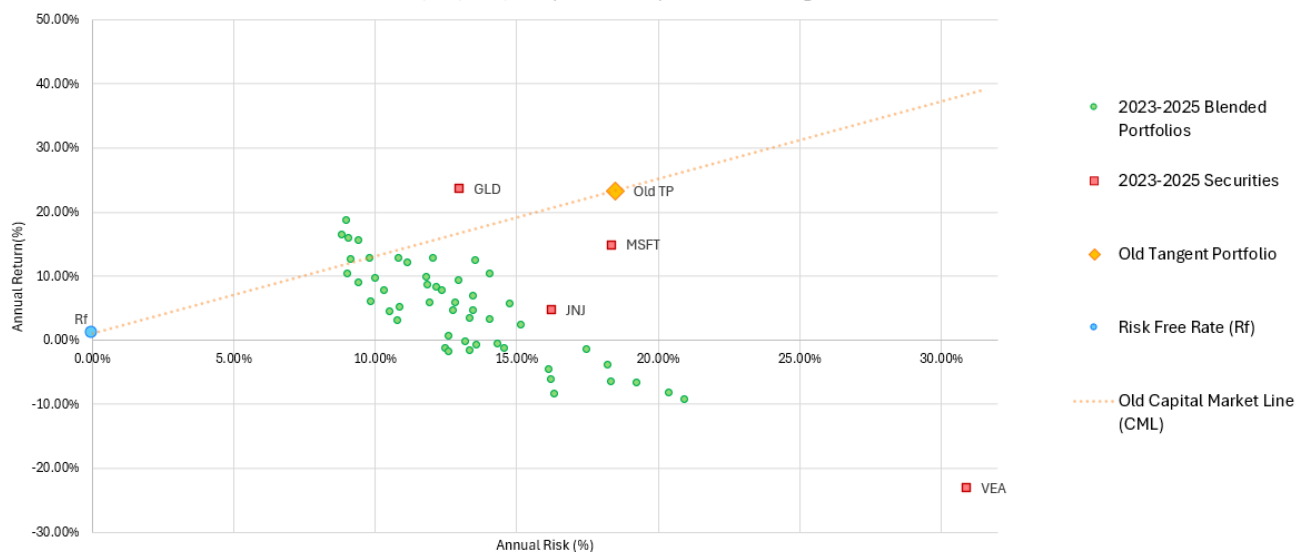
The visual representation of the Old Capital Market Line (CML) in conjunction with the 2023–2025 realized data reveals that the Old Tangent Portfolio (Old TP) is no longer tangent to the efficient frontier. Many blended portfolios now lie above the old CML, clearly demonstrating that the old TP is no longer the optimal risk-return combination. This misalignment emphasizes the need for portfolio reoptimization in response to structural shifts in return distributions and risk preferences post-2023.

The updated CML, derived from the actual return distributions over the 2023–2025 period, would connect the new optimal tangent portfolio with the risk-free rate, and display a steeper or slope, capturing the new market conditions.

Security Market Line vs Actual Returns (2023 - 2025)



Risk vs Return of MSFT, JNJ, GLD, VEA (2023 - 2025) with the Old Tangent Portfolio and CML



This adjustment reinforces a key lesson: while CAPM and mean-variance optimization are powerful tools, their effectiveness is contingent upon regularly updating portfolio weights and return assumptions in a dynamic market.

Current Event Implications

The closing months of the dataset (January–March 2025) showed signs of volatility and downturn in equities, especially in VEA and MSFT. These coincided with the geopolitical climate surrounding Trump's Liberation Day on April 2, 2025, and subsequent tariff announcements. While the data set does not explicitly contain the closing dates of Liberation Day, it contains the economic climate leading up to it, and some interesting implications can be seen:

- VEA dropped -33.65% in Feb 2025 alone
- GLD gained +9.45% by March 2025

VEA represents international developed markets such as Canada and Europe and thus a logical explanation for its sudden downturn are the tariffs that were being planned by the Trump Administration at this time. Meanwhile, GLD saw a sharp rise during this same period, consistent with a safety behavior that investors often make when fear and volatility increase.

This transition period demonstrated a shift in investor sentiment and really highlighted the failure of a static TP-based CAPM model. A forward-looking model would likely require periodic recalibration or integration with indicators and would be much better adjusted for facing transition periods within markets, such as the one seen these last few months.

Reflections on Process and Implementation

The execution of this project comprised precision in model pipeline organization, where spreadsheet design was devised thoughtfully, return figures were documented in a well-disciplined fashion, and Excel formula order was logical. The organization required at the outset was considerable for tidy tables of monthly returns, portfolio weights, and the framework required to calculate annualized metrics, all of which was instrumental in maintaining the model interpretable and flexible as the analysis grew.

Some practical observations were made while modeling:

- **Solver Setup and Optimization:** Although the Excel Solver add-in was instrumental in building the Tangent Portfolio (TP), it first failed because of weight definitions that were not well constrained. It was subsequently realized that a weight must be defined as a residual (1 minus the sum of the others) to avoid Solver's capital misallocation. After fixing this, Solver consistently optimized the Sharpe ratio without requiring tight constraints such as short-selling bans since the assets chosen lacked extreme positions.
- **Graphical Feedback and Visual Debugging:** Charting proved to be one of the most surprisingly useful diagnostic aids. For instance, when the first TP weights were plotted on the Capital Market Line (CML), the TP was not seen to be tangent to the efficient frontier. This graphical feedback immediately indicated that something had gone awry with the solver. The charts were effectively mirrors of the logic contained within the data.
- **Beta Construction Options:** Computing beta through regression and the covariance formula brought out the value of redundancy in modeling. Although both methods produced the same values—verifying mathematically that regression and covariance are equivalent—it checked the setup and allowed intuitive understanding of both methods' mechanics. Having the ability to calculate both allowed learning at a more profound level beyond mere acceptance of one black-box outcome.

- **Chart Layering and Sequencing:** With additional features being included—security points, TP markers, trendlines, forecast overlays—chart layering and labeling needed to be managed with growing attention. Keeping the Capital Market Line from being obscured or making regression lines readable in dense scatterplots, demanded continuous graphical iteration. Communicative clarity became equally significant as computational precision.

- **Dynamic Excel Logic:** The construction of the CAPM projection engine and the conversion of betas to forward period expected returns necessitated computable, scalable formulas in modules. Determining where to input weights versus calculate them dynamically minimized human error, and the exercise overall made Excel's value not only in number crunching but also in dynamic model creation very clear.

In the end, the process's success was less a matter of the answers it provided and more a matter of the strength of the apparatus created to provide them. The meticulous design of return tables, Solver constraints, visual checks, and formula checks produced not just operational accuracy but pedagogical clarity—each tool reinforced the others, and the model overall was greater than the sum of its parts. The most valuable learning was indeed in the process of model building itself, especially how slight errors in setup could have disproportionate effects later in interpretation.

Conclusion

The project validates the theoretical foundations of the CAPM Model. It affirms CAPM's explanatory power when having full hindsight but also uncovered deficiencies in predicting under market change. The post-2023 results, driven significantly by volatility and uncertainty, demonstrated the need for dynamic models for portfolio management via a real-world setting. To improve this model, something like rolling betas relative to a given window size would be one step that would tend to reduce the extent to which CAPM become stale when making forecasts further into the future.